

Proof

Introduction to proofs

A proof is a valid argument that establishes the truth of a mathematical statement (theorem).

Direct Proofs

A **direct proof** shows that a conditional statement $p \rightarrow q$ is true by showing that if p is true, then q must also be true.

Definition 1: The integer n is *even* if there exists an integer k such that $n = 2k$, and it is *odd* if there exists an integer k such that $n = 2k + 1$.

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is an odd integer.”

Solution:

Assume that the hypothesis of this conditional statement is true, namely, assume that n is an odd integer.

Then $n = 2k + 1$, where k is an integer.

It follows that

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2m + 1, \text{ where } m = 2k^2 + 2k \text{ is an integer} \end{aligned}$$

So, n^2 is an odd integer.

Therefore, the given theorem is proved.

Example: Give a direct proof that “If m and n are both perfect squares, then mn is also a perfect square.”
(A non-negative integer a is a **perfect square** if there is a non-negative integer b such that $a = b^2$.)

Solution:

Assume that m and n are perfect squares.

Then $m = s^2$ and $n = t^2$, where s and t are non-negative integers.

It follows that

$mn = s^2 t^2 = (st)^2 = k^2$, where $k = st$ is a non-negative integer.

So, mn is perfect square.

Therefore, the given theorem is proved.

Proofs by Contraposition

Proofs by Contraposition make use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive, $\neg q \rightarrow \neg p$.

This means that the conditional statement $p \rightarrow q$ can be proved by showing that its contrapositive, $\neg q \rightarrow \neg p$, is true.

Example: Prove that “If $3n + 2$ is odd, then n is odd,” where n is an integer.

Solution:

Implication: If $3n + 2$ is odd, then n is odd.

Contrapositive: If n is even, then $3n + 2$ is even.

Assume that n is even.

Then $n = 2k$ for some integer k .

It follows that

$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2m$, where $m = 3k + 1$ is an integer.

So, $3n + 2$ is even.

As the contrapositive is true, the original implication is true.

Example: Prove that “If $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$,” where a and b are positive integers.

Solution:

Implication: If $n = ab$, then $(a \leq \sqrt{n}) \vee (b \leq \sqrt{n})$

Contrapositive: If $\neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n}))$, then $n \neq ab$

Assume that $\neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n}))$ is true.

Then

$$\begin{aligned} & \neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n})) \\ & \equiv \neg (a \leq \sqrt{n}) \wedge \neg (b \leq \sqrt{n}) \\ & \equiv (a > \sqrt{n}) \wedge (b > \sqrt{n}) \end{aligned}$$

This shows that

$$ab > \sqrt{n} \cdot \sqrt{n} = n$$

Therefore, $n \neq ab$

As the contrapositive is true, the original implication is true.

Proofs by Contradiction

The statement $(r \wedge \neg r)$ is a contradiction, where r is any proposition.

We can prove that a proposition p is true if we can show that $\neg p \rightarrow (r \wedge \neg r)$ is true.

Example: Prove that “ $\sqrt{2}$ is irrational” by giving a proof by contradiction.

Solution:

Suppose that $\sqrt{2}$ is rational. Then, $\sqrt{2} = a/b$, where a and b are integers and have no common factors, $b \neq 0$.

$$\begin{aligned}\sqrt{2} &= a/b \\ \text{or, } 2 &= a^2/b^2 \\ \text{so, } 2b^2 &= a^2\end{aligned}$$

It follows that a^2 is even and a must also be even. Then, $a = 2c$ for some integer c .

Thus,

$$\begin{aligned}2b^2 &= a^2 = 4c^2 \\ \text{so, } b^2 &= 2c^2\end{aligned}$$

This means that b^2 is even and b must be even. Then, $b = 2d$ for some integer d .

Now, $\sqrt{2} = a/b = (2c)/(2d)$

Thus, a and b have a common factor 2, which contradicts with the assumption that a and b have no common factors.

Therefore, $\sqrt{2}$ is not rational, that means, $\sqrt{2}$ is irrational.

Example: Give a proof by contradiction of the theorem “If $3n + 2$ is odd, then n is odd.”

Note: $\neg (p \rightarrow q) \equiv \neg (\neg p \vee q) \equiv (p \wedge \neg q)$

Solution:

Assume that “If $3n + 2$ is odd, then n is even.”

Then $n = 2k$ for some integer k .

It follows that

$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1), \text{ where } (3k + 1) \text{ is an integer.} \end{aligned}$$

So, $3n + 2$ is even. This contradicts the assumption that $3n + 2$ is odd.

Therefore, “If $3n + 2$ is odd, then n is even” is not true.

This implies that “If $3n + 2$ is odd, then n is odd” is true.

Inductive Proof

Mathematical Induction

Mathematical induction is used to prove statements that assert that $P(n)$ is true for all nonnegative or positive integers n , where $P(n)$ is a propositional function.

PRINCIPLE OF MATHEMATICAL INDUCTION

To prove that $P(n)$ is true for all nonnegative or positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(0)$ or $P(1)$ (which one is the first element of the domain) is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all nonnegative or positive integers k .

To complete the inductive step of the proof, we assume that $P(k)$ is true for an arbitrary nonnegative or positive integer k and show that under this assumption, $P(k + 1)$ must also be true. The assumption that $P(k)$ is true is called the **inductive hypothesis**.

Once we complete both steps in a proof by mathematical induction, we have shown that $P(n)$ is true for all nonnegative or positive integers, that is, we have shown that $\forall n P(n)$ is true where the quantification is over the set of nonnegative or positive integers.

Example: Show that if n is a positive integer, then $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Solution:

Let $P(n)$: $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

BASIS STEP: Let $n = 1$

LHS = 1 and RHS = $\frac{1(1+1)}{2} = 1$. As LHS = RHS, $P(1)$ is true.

INDUCTIVE STEP: Assume that $P(k)$ holds for an arbitrary positive integer k . That is, we assume that
 $P(k)$: $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$ (1)

Under this assumption, it must be shown that $P(k + 1)$ is true.

Adding $(k + 1)$ at both the sides of (1)

$$1 + 2 + \cdots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2} = \frac{(k+1)[(k+1)+1]}{2} = P(k + 1)$$

Therefore, $P(k) \rightarrow P(k + 1)$ is true.

Since $P(1)$ is true and $P(k) \rightarrow P(k + 1)$ is true for all positive integers k , $P(n)$ is true for all positive integers n .

Example: Prove that the sum of the first n odd positive integers is n^2 .

Solution:

Let $P(n)$: $1 + 3 + 5 + \dots + (2n - 1) = n^2$.

BASIS STEP: Let $n = 1$

LHS = 1 and RHS = $1^2 = 1$. As LHS = RHS, $P(1)$ is true.

INDUCTIVE STEP: Assume that $P(k)$ holds for an arbitrary positive integer k . That is, we assume that $P(k)$: $1 + 3 + 5 + \dots + (2k - 1) = k^2$. (1)

Under this assumption, it must be shown that $P(k + 1)$ is true.

Adding $[2(k + 1) - 1]$ at both the sides of (1)

$$1 + 3 + 5 + \dots + (2k - 1) + [2(k + 1) - 1] = k^2 + [2(k + 1) - 1] = k^2 + 2k + 2 - 1 = k^2 + 2k + 1 = (k + 1)^2 = P(k + 1)$$

Therefore, $P(k) \rightarrow P(k + 1)$ is true.

Since $P(1)$ is true and $P(k) \rightarrow P(k + 1)$ is true for all positive integers k , $P(n)$ is true for all positive integers n .

Example: Use mathematical induction to show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n .

Solution:

Let $P(n)$: $1 + 2 + 2^2 + \dots + 2^n = 2^0 + 2^1 + 2^2 + \dots + 2^n = 2^{n+1} - 1$.

BASIS STEP: Let $n = 0$

LHS = $2^0 = 1$ and RHS = $2^{0+1} - 1 = 1$. As LHS = RHS, $P(0)$ is true.

INDUCTIVE STEP: Assume that $P(k)$ is true, that is, we assume that

$$P(k): 2^0 + 2^1 + 2^2 + \dots + 2^k = 2^{k+1} - 1 \quad (1)$$

Under this assumption, it must be shown that $P(k + 1)$ is true.

Adding 2^{k+1} at both the sides of (1)

$$2^0 + 2^1 + 2^2 + \dots + 2^k + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{(k+1)+1} - 1 = P(k + 1)$$

Therefore, $P(k) \rightarrow P(k + 1)$ is true.

Since $P(0)$ is true and $P(k) \rightarrow P(k + 1)$ is true for all nonnegative integers k , $P(n)$ is true for all nonnegative integers n .